Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of matrix metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers

GLENN P. LADWIG, JD, MS[a]; MARTIN C. ROBSON, MD[b]; RAN LIU, MD[a]; M. ANN KUHN, MD[b]; DAVID F. MUIR, PhD[c]; GREGORY S. SCHULTZ, PhD[a]

Previous analyses of fluids collected from chronic, nonhealing wounds found elevated levels of inflammatory cytokines, elevated levels of proteinases, and low levels of growth factor activity compared with fluids collected from acute, healing wounds. This led to the general hypothesis that chronic inflammation in acute wounds produces elevated levels of proteinases that destroy essential growth factors, receptors, and extracellular matrix proteins, which ultimately prevent wounds from healing. To test this hypothesis further, pro- and activated matrix metalloproteinases (MMP-2 and MMP-9), tissue inhibitors of metalloproteinases (TIMP-1 and TIMP-2), and the ratios of MMPs/TIMPs were assayed in fluids and biopsies collected from 56 patients with chronic pressure ulcers. Specimens included ulcers treated for 0, 10, and 36 days with conventional therapy or with exogenous cytokine therapies. Quantitative assay data were correlated with the amount of healing. The average MMP-9/TIMP-1 ratio in fluids from 56 ulcers decreased significantly as the chronic pressure ulcers healed. Furthermore, the average MMP-9/TIMP-1 ratio was significantly lower for fluids collected on day 0 from wounds that ultimately healed well (≥85% reduction in initial wound volume) compared with wounds that healed poorly (<50% wound volume reduction). These data show that the ratio of MMP-9/TIMP-1 levels is a predictor of healing in pressure ulcers and they provide additional support for the hypothesis that high levels of MMP activity and low levels of MMP inhibitor impair wound healing in chronic pressure ulcers. **(WOUND REP REG 2002;10:26–37)**

Healing of skin wounds normally occurs in a predictable sequence of phases including hemostasis, inflammation, mitosis, angiogenesis, and synthesis of extracellular matrix (ECM) followed by remodeling of the scar matrix.[1–3] These processes are regulated by numerous molecules including growth factors, cytokines, proteinases, and their inhibitors. Chronic failure of an acute wound to progress through the phases of healing is probably due to interference with the normal interactions of these molecules. Previous studies that analyzed fluids collected from chronic wounds found elevated levels of inflammatory cytokines, elevated levels of proteinases, and low levels of growth factor activity compared with acute, healing wounds.[4–20] These

APMA	p-Aminophenyl mercuric acetate
bFGF	basic Fibroblast growth factor
ECM	Extracellular matrix
ELISA	Enzyme-linked immunosorbent assay
GM-CSF	Granulocyte macrophage-colony stimulating factor
MMP	Matrix metalloproteinase
TIMP-1	Tissue inhibitor of metalloproteinase-1
TNF-α	Tumor necrosis factor-α

From the Institute for Wound Research[a], Department of Obstetrics and Gynecology and Department of Neuroscience[c], University of Florida, Gainesville; and the Institute for Tissue Regeneration, Repair and Rehabilitation[b], Bay Pines VAMC, Bay Pines, Florida.

Reprint Requests: Gregory S. Schultz, PhD, Institute for Wound Research, Department of Obstetrics and Gynecology, 1600 SW Archer Road, Gainesville, FL 32610-0294. Fax: (352) 392-6994; Email: schultzg@obgyn.ufl.edu.

Copyright © 2002 by the Wound Healing Society.

ISSN: 1067-1927 $15.00 + 0

observations led to the hypothesis that chronic wounds develop because of prolonged inflammation in acute wounds which produces elevated levels of proteinases that destroy growth factors, receptors, and ECM proteins that are essential for healing.[21,22] If this hypothesis is correct, it follows that elevated levels of inflammatory cytokines and proteinases should decrease as chronic wounds begin to heal. This study examined levels of matrix metalloproteinase (MMP)-2, MMP-9 and tissue inhibitor of metalloproteinase-1 (TIMP-1) in sequential wound fluids and biopsies collected from 56 patients with chronic pressure ulcers and correlated the levels of these MMPs and inhibitor with the extent of ulcer healing.

MATERIALS AND METHODS

Wound fluid and tissue samples were collected from patients enrolled in a four-arm, blinded, prospective, randomized placebo-controlled pressure ulcer clinical trial comparing topical administration of granulocyte macrophage-colony stimulating factor (GM-CSF), basic fibroblast growth factor (bFGF), and sequential treatment with GM-CSF followed by bFGF over a 35-day period. Details of the clinical trial and results have been published previously.[23] Briefly, ulcers were between 10 and 200 cm^3 and of least 8 weeks duration. Wound biopsies and fluid samples were taken from the ulcers at day 0 (pretreatment), day 10, and day 36. Measurements of the pressure ulcer volume were performed on day 0 and weekly for 5 weeks using planimetry of the ulcer opening and volume determination using alginate molds.[24] Five previous trials of 130 patients with pressure ulcers reported a mean healing response of a 70% ± 2% decrease in ulcer volume over a 4-week period for patients treated with placebo.[25] Therefore, in this study, a response of at least an 85% decrease in wound volume over 35 days was chosen as indicative of the good healers, intermediate healers were defined as a 50–85% decrease in volume, and poor healers were defined as a decrease in wound volume of less than 50%.

Collection and processing of wound fluids and biopsies

Wound fluids were collected from the ulcers using porous, inert hydrophilic dextranomer beads.[26] Briefly, beads were placed in each pressure ulcer forming a layer approximately ¼ inch thick. The ulcer was dressed with an occlusive dressing and covered with an adherent elastic wrap. The beads remained in the ulcers for 24 hours then the beads were collected, and 1 gram of the saturated beads was mixed with 1 ml of 100 mM sodium phosphate, pH 7.4 for 12 hours in a vertical shaker at 4 °C to elute the wound fluid protein from the dextran beads. The beads were then centrifuged at 1000 × g for 5 minutes at 4 °C and the supernatant solution were stored at −80 °C until analyzed.

A 4-mm punch biopsy of tissues was collected at either the center or the edge of the pressure ulcer from 13 patients and stored at −80 °C. Frozen punch biopsies were weighed, cut into small pieces, and then homogenized using a 1-ml frosted glass-on-glass homogenizer (Wheaton, Milville, NJ) in buffer (20 mM Tris, 0.1% Triton X-100, pH 7.4) at a ratio of 100 mg tissue/ml of homogenizing buffer. The homogenates were centrifuged at 14,000 × g for 5 minutes to remove particulate matter, and the supernatant was stored −80 °C until analyzed.

Quantitative gelatin zymography

Levels of the pro and activated forms of the two gelatinases, MMP-2 (gelatinase A) and MMP-9 (gelatinase B), were measured in fluids and homogenized biopsies from pressure ulcers using quantitative gelatin zymography.[10,27] Briefly, 15 µl of appropriately diluted wound fluid or biopsy homogenate supernatant were mixed with an equal volume of sample buffer (63 mM Tris-HCl, pH 6.8, 10% glycerol, 2% sodium dodecyl sulfate and 0.0025% bromophenol blue; Novex, San Diego, CA) and incubated at room temperature for 10 minutes. Twenty µl of the equilibrated sample were loaded into a well of a 15-well precast gelatin zymogram gel (Novex, San Diego, CA). To measure levels of pro and activated MMP-9, samples were electrophoresed at 4 °C at 95 V until the bromophenol blue tracking dye had migrated through the stacking gel and then were electrophoresed at 125 V until the tracking dye reached the bottom of the gel (approximately 2.5 hours). To measure levels of pro and activated MMP-2, gels were run an additional hour after the tracking dye reached the bottom of the gel (approximately 3.5 hours). Gels were then immersed in renaturing buffer (2.7% Triton X-100, w/v) and placed on a rotary shaker at 30 r.p.m. for 30 minutes at 37 °C. The gels were then placed in developing buffer (50 mM Tris-HCl, 200 mM NaCl, 5 mM $CaCl_2$, 0.2% Brij 35, w/v) on a rotary shaker at 37 °C for 24 hours to allow the MMPs to digest the gelatin substrate. After digestion, the gels were stained with Coomassie Rapid Stain (Diversified Biotech, Boston, MA) and destained with 12.5% trichloroacetic acid. Gels were photographed with a Kodak KD120 digital camera (Eastman, NY) and the relative pixel density of each band was measured using Kodak Digital Science image analysis software.[28,29] Pre-stained molecular weight standards (Novex, San Diego, CA) and one sample of pro and activated MMP-9 and MMP-2 were run on each gel.

Standard curves for pro and activated forms of MMP-2 and MMP-9

Levels of pro and activated MMP-2 and MMP-9 in samples were calculated from standard curves generated with recombinant pro and activated MMP-2 and MMP-9 (Oncogene Research Products, Cambridge, MA). Briefly, serial twofold dilutions of the pro and activated forms of MMP-2 and MMP-9 were subjected to gelatin zymography as described above. Bands were digitized and the amount of recombinant protein was plotted versus the band intensity. Standard curves (Figure 1, panels A–D) were generated for each enzyme form. The ranges of MMP standards and correlation coefficients were: 39–10,000 pg of pro MMP-2 ($r^2 = 0.98$); 39–5000 pg of activated MMP-2 ($r^2 = 0.99$); 5–1250 pg of pro MMP-9 ($r^2 = 0.98$); and 5–625 pg of activated MMP-9 ($r^2 = 0.99$). Levels of pro and activated MMP-2 and MMP-9 were expressed as ng of MMP/ml of wound fluid or as ng of MMP/gm of biopsy.

It was observed that there were relatively higher concentrations of MMP-9 than MMP-2 in most wound fluid samples. Therefore, different dilutions of the samples were made to generate band intensities for the MMPs that fell in the optimal ranges of their standard curves. Typically, wound fluid samples were diluted 150-fold in phosphate buffer to measure levels of pro and activated MMP-2, while wound fluid samples were diluted 500-fold to measure levels of pro and activated MMP-9.

Gelatin zymography of samples treated with an inhibitor and an activator of MMPs

To confirm that the bands detected in the gelatin zymograms were produced by MMP activities, replicate gelatin zymogram gels were developed in the presence of the MMMP inhibitor 1,10-phenanthroline (5 mM). In addition, an experiment was performed to activate the latent MMP-9 recombinant standard using the organomercurial reagent, p-aminophenyl mercuric acetate (APMA; Sigma Chemical Co., St. Louis, MO). APMA was added to native samples to a final concentration of 1 mM and incubated for 4 hours at 37 °C. Then the sample was applied to the gelatin zymogram.[30]

Western blot analysis of MMP-2 and MMP-9

To further characterize the bands observed by zymography, Western blots were performed using modifications of a previously reported technique.[14] Briefly, MMP standards (25 ng), wound fluids, and biopsy homogenates samples (10 μl of undiluted samples) were electroblotted from gels

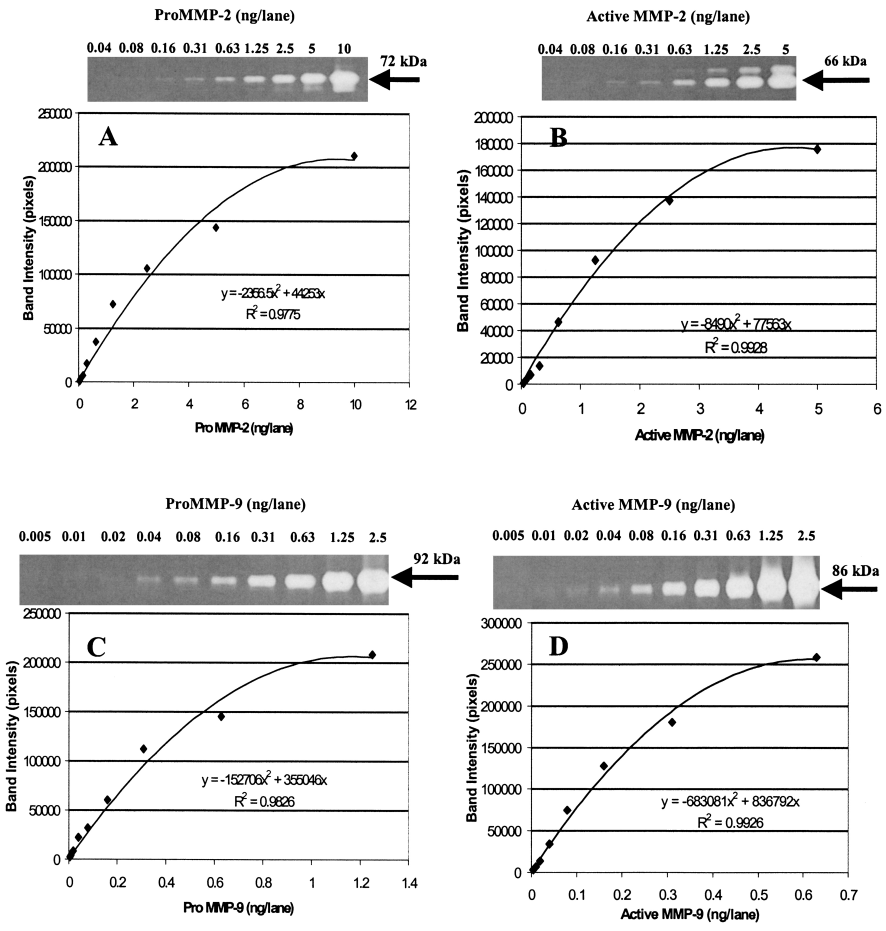


FIGURE 1. Quantitative gelatin zymography for pro and activated MMP-2 and MMP-9. Standard curves were generated for pro MMP-2 (panel A), activated MMP-2 (panel B), pro MMP-9 (panel C), and activated MMP-9 (panel D) using the amounts of recombinant MMPs indicated above each lane of the gelatin zymograms. The quadratic equations and correlation coefficients are shown for each curve.

to nitrocellulose membranes (Millipore, Bedford, MA), rinsed briefly in de-ionized water, and fixed in 25% isopropanol/10% acetic acid/65% water for 30 minutes at room temperature with agitation. After rinsing in water for 10 minutes, the blot was equilibrated for 10 minutes in washing buffer (50 mM Tris-HCl, pH 7.4, 1.5% NaCl and 0.1% Triton X-100) then blocked for 1 hour in washing buffer containing 5% (w/v) nonfat dry milk. The blot was incubated for 2 hours with the primary antibodies, which were diluted to 1 μg/ml in blocking buffer. Primary antibodies were mouse monoclonal anti-human MMP-2 and anti-human MMP-9, which bind both the pro and activated forms of the MMPs (Oncogene Research Products, Cambridge, MA). After incubation with the primary antibodies, the blots were washed and then incubated with a peroxidase-conjugated secondary antibody for 2 hours (porcine anti-mouse IgG-horseradish peroxidase, IgG-HRP, Dako, Carpinteria, CA) at a 1:1000 dilution with blocking buffer. The blot was washed and subjected to a chemiluminescence detection system (Pierce, Rockford, IL) as described by the manufacturer's protocol.

TIMP-1 enzyme-linked immunosorbent assay

Levels of TIMP-1 were measured in wound fluids and tissue biopsies using a commercially available enzyme-linked immunosorbent assay (ELISA; Amersham Pharmacia Biotech, Piscataway, NJ). The assay has a sensitivity of 1.25 ng/ml and recognizes total human TIMP-1, i.e., free TIMP-1 and TIMP-1 in complexes with MMP-1, MMP-2, MMP-3, MMP-9 and pro MMP-9. The TIMP-1 ELISA does not cross-react with TIMP-2, TIMP-3, or TIMP-4. Levels of TIMPs in samples were expressed as ng/ml of wound fluid and ng/gm of biopsy tissue.

Statistical analysis

The data were categorized into treatment groups and clinical response groups. Because there were insufficient frozen biopsy samples for a statistical comparison, only gelatinolytic profiles between wound fluids and biopsy homogenates were qualitatively compared. All zymogram and ELISA data were statistically analyzed using a multivariate analysis of variance (MANOVA) with Statistica software (StatSoft Inc., Tulsa, OK) to determine significant differences between treatment groups and clinical response groups.

All sequential data obtained on gelatinase activity, TIMP-1 activity, and ulcer measurements were evaluated for possible correlation using the Spearman Rank Order Correlation. With this test, pairs of variables with positive correlation coefficients and p-values <0.05 tend to increase together. For pairs of variables with negative correlation coefficients and p-values <0.05, one variable tends to decrease while the other increases. The Spearman Rank Order Correlation was performed on these data comparing the percent decrease in wound volume and relative change in gelatinase and TIMP-1 activity. The data were then categorized by treatment group and clinical response group and another Spearman Rank Order Correlation was performed. In addition, a two-sample t-test assuming unequal variances was performed. Statistical significance was accepted at $p < 0.05$.

RESULTS

The standard curves for pro MMP-2, activated MMP-2, pro MMP-9, and activated MMP-9 are shown in Figure 1, panels A–D, respectively. Separation of each of the lysis bands corresponding to the pro and activated MMP-2 and MMP-9 standards was sufficient to exclude contributions from other forms of MMPs. Similar to previous reports, the integrated densities of the lysis bands for the four MMP standards are nearly linear over a substantial portion of the concentration range for the standard MMPs.[10,29] However, at high levels of the MMP standards, the intensities of the lysis bands began to generate a curvilinear plot. Quadratic equations were used to describe the best fit curve to the data with correlation coefficients (r^2) ≥ 0.98 for all standard curves.

Figure 2, panels A and B show the relative sensitivities of the gelatin zymograms to the pro and activated forms of MMP-2 and MMP-9. As is evident by the different x-intercepts, as well as the slopes of the lines, the gelatin zymography was approximately eight times more sensitive in detecting MMP-9 than MMP-2. Stated differently, the limit of detection (sensitivity) for the pro and activated forms of MMP-9 was approximately eight times lower than the limit of detection for the pro and activated forms of MMP-2.

MMP levels in wound fluid and biopsies

The greater sensitivity of gelatin zymograms for MMP-9 compared with MMP-2 produces an important, and potentially misleading, visual effect when samples of wound fluids and biopsies are analyzed. For example, visual evaluation of the gelatinolytic band patterns observed in Figure 3, panel A, would suggest that biopsy samples may contain slightly higher amounts of MMP-9 than MMP-2 proteins because the band intensities and areas appear slightly larger for MMP-9 than for MMP-2. However, as shown in Figure 4, panel A, the average amount of pro MMP-2 protein is approximately 1.5-fold higher than the average amount of pro MMP-9 in biopsies, and the visual impression of the relative levels of MMPs is incorrect owing to the greater sensitivity of gelatin zymograms to MMP-9 gelatinolytic activity.

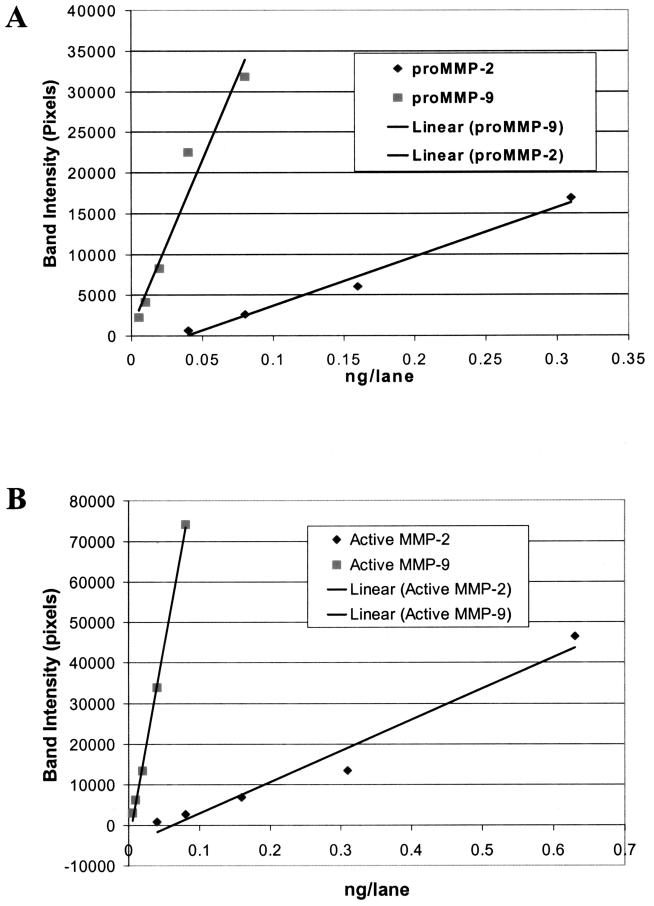


FIGURE 2. Comparison of the sensitivities of gelatin zymography between MMP-2 and MMP-9. The data generated by the standard curves presented in Figure 1 for pro MMP-2 and pro MMP-9 are plotted in panel A, and the data for activated MMP-2 and activated MMP-9 are plotted in panel B. As indicated by the slopes and intercepts, gelatin zymography is approximately eightfold more sensitive to MMP-9 than to MMP-2.

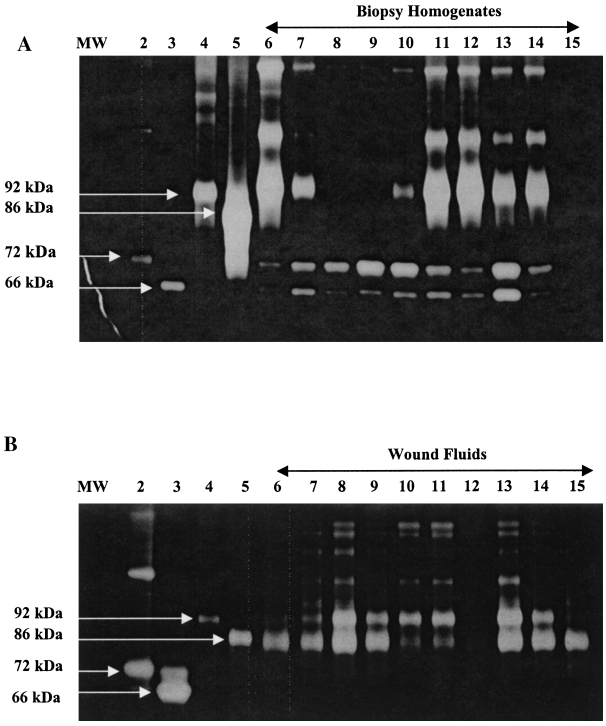


FIGURE 3. Representative gelatin zymograms of biopsy homogenates and wound fluids. Gelatin zymograms containing representative samples of biopsies (panel A) and wound fluids (panel B) are presented. In both gels, lane 1 contained molecu lar weight standards, and lanes 2 through 5 contained purified pro and activated MMP-2 (72 kDa, 66 kDa) and pro and activated MMP-9 (92 kDa, 86 kDa) standards, respectively. Lanes 6 through 15 contained biopsy homogenates (panel A) or wound fluids (panel B). Homogenates of biopsies (panel A) consistently produced both pro and activated MMP-2 and pro MMP-9 bands. In contrast, wound fluids (panel B) produced MMP-9 bands but no detectable MMP-2 bands at this dilution.

Visual evaluation of the gelatinolytic band patterns observed in Figure 3, panel B, suggests that samples of wound fluids contain abundant amounts of pro MMP-9 and activated MMP-9 and have extremely low or undetectable levels of pro MMP-2 and activated MMP-2. However, as shown in Figure 4, panel B, because of the greater sensitivity of gelatin zymograms for MMP-9 gelatinolytic activity, wound fluid samples actually contain only about four times more MMP-9 than MMP-2 when the levels of MMPs are calculated using the standard curves and appropriate dilutions of wound fluids are analyzed.

As shown in Figure 3, panel A, 9 of 10 homogenates of wound biopsies (lanes 6–15) contained bands that comigrated at 72 kDa with the pro MMP-2 standard and at 66 kDa with the activated MMP-2 standard. Furthermore, the relative intensity of the pro MMP-2 band was greater than the intensity of the activated MMP-2 band in eight of the nine homogenates. The trends that were observed in these 10 representative biopsy homogenates were repeated when all 37 biopsy samples were analyzed. For example, a band corresponding to pro-MMP-2 was observed in 35 of the 37 biopsies (95%) and a band corresponding to activated MMP-2 was observed in 34 of 37 biopsies (92%). In addition, the pro-MMP-2 band was more intense than the activated MMP-2 band by at least 1.5-fold in 28 of the 35 biopsies (80%) that had MMP-2 bands. As shown in panel A of Figure 4, the mean level in wound fluids of pro MMP-2 (90,000 ng/ml ± 11,000 ng/ml) was 4.2-fold higher than the level of activated MMP-2 (21,000 ng/ml ± 3000 ng/ml) ($p = 0.03$).

In contrast to the very high percentage of biopsies containing detectable pro MMP-2 or activated MMP-2 bands, none of 10 representative wound fluid samples contained detectable pro MMP-2 or activated MMP-2 bands when 5 μl of 500-fold diluted wound fluids were assayed (Figure 3, panel B, lanes 6–15). This observation also was confirmed when 5 μl of 200-fold dilutions of wound fluids

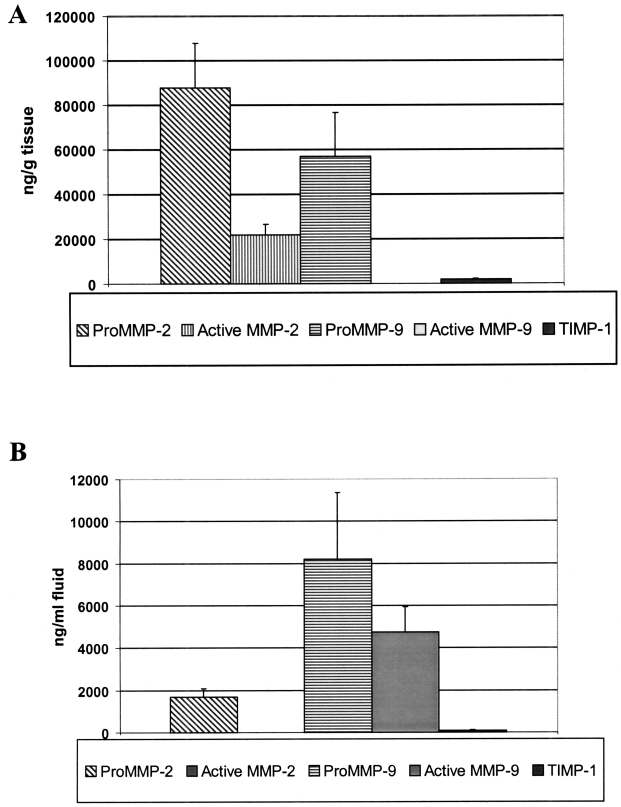


FIGURE 4. MMP-2, MMP-9 and TIMP-1 levels in biopsy homogenates and wound fluids at day 0. Levels of pro and activated MMP-2, MMP-9 and TIMP-1 (mean ± SEM) were measured in biopsies (panel A) and wound fluids (panel B) using quantitative gelatin zymography and enzyme-linked immunosorbent assay (ELISA). Biopsies typically contained pro MMP-2, activated MMP-2, and pro MMP-9 but not activated MMP-9. In contrast, wound fluids typically contained pro MMP-2, pro MMP-9, and activated MMP-9 but not activated MMP-2. Both biopsies and wound fluids contained TIMP-1.

were assayed. Only 20 of 125 (16%) wound fluid samples had a detectable pro MMP-2 band, and only 4 of the 20 fluids (3% overall) also had a detectable activated MMP-2 band. Also, the activated MMP-2 band was less than half the intensity of the pro MMP-2 band in the four wound fluid samples that had both pro and activated MMP-2 bands. As shown in panel B of Figure 4, the mean level of pro MMP-2 in biopsies was 1800 ± 200 ng/ml and the mean level of activated MMP-2 was 100 ± 50 ng/ml) ($p = 0.03$).

Analysis of the pro MMP-9 and activated MMP-9 bands in biopsies also revealed interesting general patterns. As seen in Figure 3, panel B, 9 of 10 wound fluids (lanes 6–15) contained a clearly identifiable band that migrated at 86 kDa, which coincided with the migration of the activated MMP-9 standard (lane 5). Furthermore, the intensity of the activated MMP-9 band was greater than or equal to the intensity of the pro MMP-9 band in seven of the nine wound fluid samples that had MMP-9 bands. As shown in panel A of Figure 4, the mean level of pro MMP-9 in biopsies was 55,000 ± 20,000 ng/gm and the mean level of activated MMP-9 was 800 ± 300 ng/gm) ($p = 0.03$).

Extending these trends to 125 wound fluid samples, 114 (91%) of the wound fluids contained a detectable activated MMP-9 band, and 89 (71%) contained a detectable pro MMP-9 band when 5 μl of 200-fold dilutions of wound fluids were assayed. Also, the relative intensity of the activated MMP-9 band was equal to or greater than the intensity of the pro MMP-9 band in 94 of the 125 (75%) wound fluid samples. Finally, none of the 125 wound fluids had only a pro MMP-9 band and no detectable activated MMP-9 band. As shown in panel B of Figure 4, the mean level of pro MMP-9 in ulcer fluids was 8100 ± 3000 ng/ml and the mean level of activated MMP-9 was 4800 ± 1300 ng/ml) ($p = 0.03$).

These data lead to the general conclusions that in chronic wound fluids there is relatively little pro MMP-2 or activated MMP-2 compared with levels of MMP-9, and most of the MMP-9 in wound fluid samples is activated MMP-9. This implies that pro MMP-9 molecules are rapidly converted to activated MMP-9 in the chronic wound environment as opposed to being retained or complexed as an inactivated zymogen.

Inhibition by 1, 10-phenanthroline confirm MMP activities

As shown in Figure 5, the pro MMP-9 standard (lane 1) migrated at the predicted 92-kDa size and the activated MMP-9 standard (lane 2) migrated at the expected 86-kDa size. Lane 3 contains biopsy, and lanes 4 and 5 contain wound fluids. The typical gelatinolytic patterns were generated showing substantial levels of MMP-2 and MMP-9 in the biopsy sample and only detectable levels of pro and activated MMP-9 in the wound fluid samples.

As shown in lanes 7–9 of Figure 5, addition of 1,10-phenanthroline (a known inhibitor of MMPs) to the zymography incubation buffer resulted in complete inhibition of all gelatinase activities detected in the biopsy sample from lane 3 and the wound fluid samples from lanes 4 and 5. These data confirm that the proteinase bands detected in the wound fluids and biopsies were attributed to MMP activities.

Western blot of wound samples identifies MMP-2 and MMP-9

Immunoblots were performed to further correlate the gelatinolytic bands observed by zymography with the MMP-2 and MMP-9 standards. As shown in panel A of Figure 6, 25 ng of pro MMP-2 standard (72 kDa, lane 2) and 25 ng of activated MMP-2 standard (66 kDa, lane 3) were both readily detected by the anti-MMP-2 antibody. The biopsy sample (lane 5) contained two immunoreactive

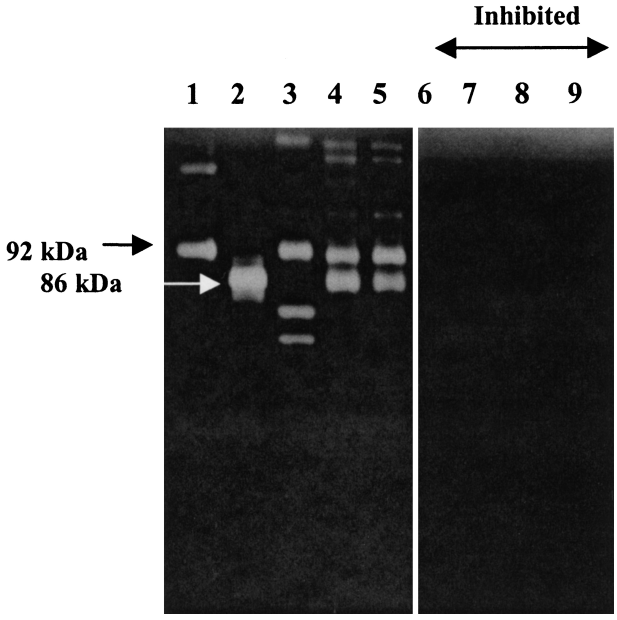


FIGURE 5. Gelatin zymogram of activated and inhibited biopsy and wound fluid samples. Gelatin zymography was performed on samples of recombinant pro MMP-9 (lane 1) and activated MMP-9 (lane 2). Lane 3 contained a biopsy homogenate, and lanes 4 and 5 contained two different wound fluid samples. Lane 6 contained pro and activated MMP-9, and lanes 7 through 9 contained the same samples as lanes 3 through 5 but were incubated with 5 mM 1,10-phenanthroline during development. The gelatinolytic bands corresponding to pro MMP-9 and activated MMP-9 were present in the biopsy and wound fluid samples (lanes 3–5). Incubation with 1,10-phenanthroline during development inhibited gelatinase activity, indicating that the bands in lanes 3–5 were produced by MMPs.

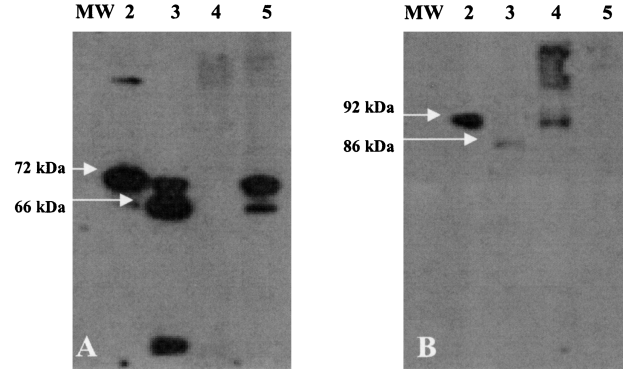


FIGURE 6. Western blots of MMP-2 and MMP-9 in wound fluid and biopsy samples. Immunoreactive MMP-2 (panel A) and MMP-9 (panel B) were visualized in biopsy and wound fluid samples by Western blotting. Panel A contained recombinant pro MMP-2 (lane 2), activated MMP-2 (lane 3), and panel B contained pro MMP-9 (lane 2) and activated MMP-9 (lane 3). A wound fluid (lane 4), a biopsy (lane 5), and molecular weight standards (left lanes) were on both panels A and B. Immunoreactive pro MMP-2 and activated MMP-2 were readily detected in the biopsy sample (lane 5, panel A) but not in the wound fluid sample (lane 4). In panel B, recombinant pro MMP-9 (lane B) was detected more readily than activated MMP-9 (lane 5). Pro MMP-9 and higher-molecular-weight bands were readily detected In the wound fluid sample (lane 4) but not in the biopsy sample (lane 5).

bands that comigrated with the pro MMP-2 and activated MMP-2 standards. This agrees with the results obtained from the gelatin zymograms, which consistently detected both pro MMP-2 and activated MMP-2 bands in biopsy samples. Very faint immunoreactive MMP-2 bands were detected in the wound fluid sample (lane 4). This is also consistent with the observation from gelatin zymograms that wound fluids typically contain extremely low levels of MMP-2 compared with biopsy samples.

As seen in panel B of Figure 6, 25 ng of pro MMP-9 standard (lane 2; 92 kDa) was readily detected in the Western blot. However, 25 ng of activated MMP-9 (lane 3; 86 kDa) produced only a weak band. This suggests that the monoclonal antibody detects pro MMP-9 protein much more readily than activated MMP-9 protein. A weak band corresponding to pro MMP-9 protein was generated in the wound fluid sample (lane 4). In addition, diffuse higher-molecular-weight bands of immunoreactive MMP-9 protein were evident in lane 4. Similar higher-molecular-weight gelatinolytic bands were visible by zymography as well (Figure 3, panels A and B), which may consist of aggregates of MMP-9 protein or TIMP/MMP-9 complexes that are recognized by the MMP-9 antibody. Very faint MMP-9 immunoreactive bands were detected in the biopsy sample (lane 5), which probably reflects the relatively low levels of MMP-9 protein detected in biopsy samples by gelatin zymography and the relatively poor recognition of activated MMP-9 by this antibody.

Relationship between MMP and TIMP levels and cytokine treatment

There were no statistically significant differences detected between the average levels of MMP-2, MMP-9, and TIMP-1, on any of the three sampling days and any of the cytokine treatment arms (GM-CSF, bFGF, or sequential treatment with GM-CSF followed by bFGF). However, MMP and TIMP analysis based on clinical response did produce significant differences, as discussed below.

Patients can be clustered into three groups based on their healing response

The clinical description of the percentage of wound closure for each of the 61 patients at the end of the 36-day study has been reported previously.[23] Samples from 56 of the 61 patients were used in the present study. Based on the extent of wound closure, the patients were clustered into three groups, regardless of cytokine treatment. As shown in Figures 7, 13 patients were designated as good healers

(their pressure ulcer healed >85%), 37 patients were designated as intermediate healers because their pressure ulcers healed <85% but >45%, and eight patients were designated as poor healers because their ulcer healed <45%.

MMP-9 levels in wound fluids were significantly higher at day 0 in patients who healed poorly

As shown in Figure 8, levels of activated MMP-9 in wound fluids were significantly ($p < 0.05$) higher at day 0 (pretreatment) in those patients who healed poorly compared with those patients who healed well (good healers). On day 10, both pro MMP-9 and activated MMP-9 exhibited a significant (Spearman's Rank Correlation, $p < 0.05$) inverse relationship with percent closure of the wound. On day 36, only activated MMP-9 exhibited an inverse relationship with percent wound closure ($p < 0.05$). Overall, the general impression is that levels of activated MMP-9 tend to decrease as healing proceeds on days 10 and 36 for intermediate and poor healers, while they remain at a constant low level in fluids of ulcers that healed well.

As shown in Figure 9, the difference in levels of pro-MMP-2 in wound fluids at day 0 between good healers and poor healers approached statistical significance ($P = 0.077$). The levels of MMP-2 did not progressively decrease as healing proceeded on days 10 and 36 but remained nearly constant. This might indicate that the expression of MMP-2 in skin is more constitutive and is regulated independently and to a lesser extent than MMP-9.[19,31]

TIMP-1 levels in wound fluids were lower at day 0 in patients who healed poorly

As seen in Figure 10, levels of TIMP-1 at day 0 were significantly lower in patients who healed poorly compared with patients who healed well ($p = 0.05$). Also, the levels of

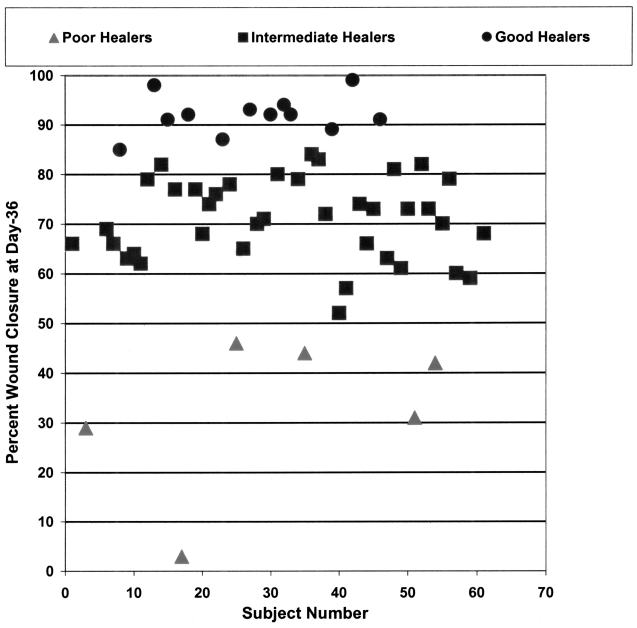


FIGURE 7. Distribution of clinical outcomes of patients treated in the companion study. A clinical response rate of 85% wound closure or greater after 35 days was chosen as indicative of a ''good healer.'' A clinical response rate of less than 50% wound closure was indicative of a ''poor healer.'' A clinical response rate of between 51% and 84% wound closure was indicative of an ''intermediate healer.''

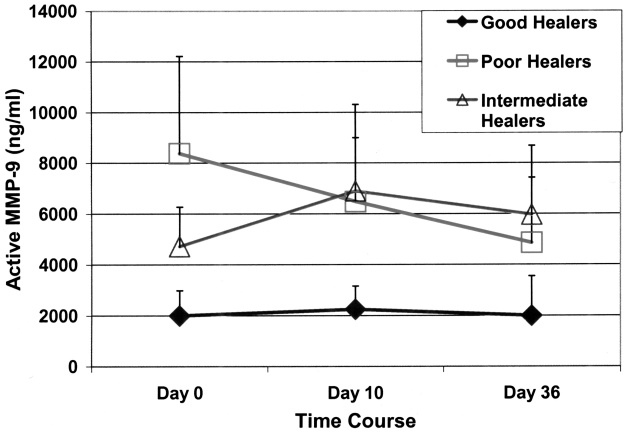


FIGURE 8. Levels of activated MMP-9 in wound fluids categorized by clinical response group. The average level of activated MMP-9 in wound fluids from good healers was significantly lower than levels of activated MMP-9 in poor responders and poor healers at day 0 (pretreatment) as measured by quantitative zymography ($p < 0.05$). Values are the mean ± SEM. Good healers: $n = 12$; poor healers: $n = 8$; *intermediate healers:* $n = 36$).

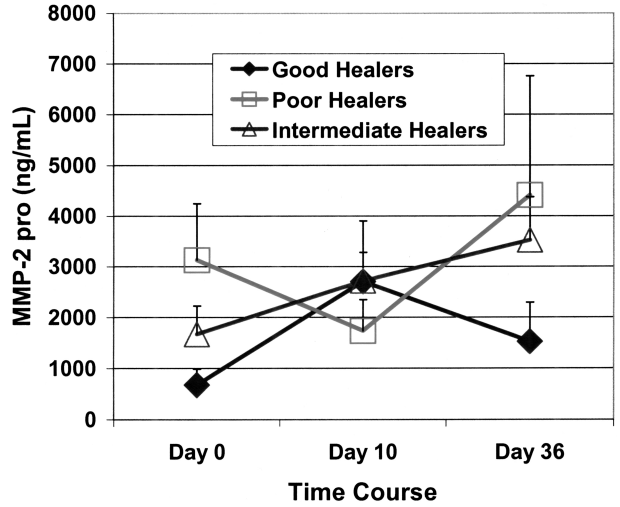


FIGURE 9. Levels of pro MMP-2 in wound fluids categorized by clinical response group. There was a trend toward lower levels of pro MMP-2 in wound fluids from good healers compared with poor healers at day 0 (pretreatment) as measured by quantitative zymography ($p < 0.08$). Values are the mean ± SEM. (Good healers: $n = 12$; poor healers: $n = 8$; intermediate healers: $n = 36$).

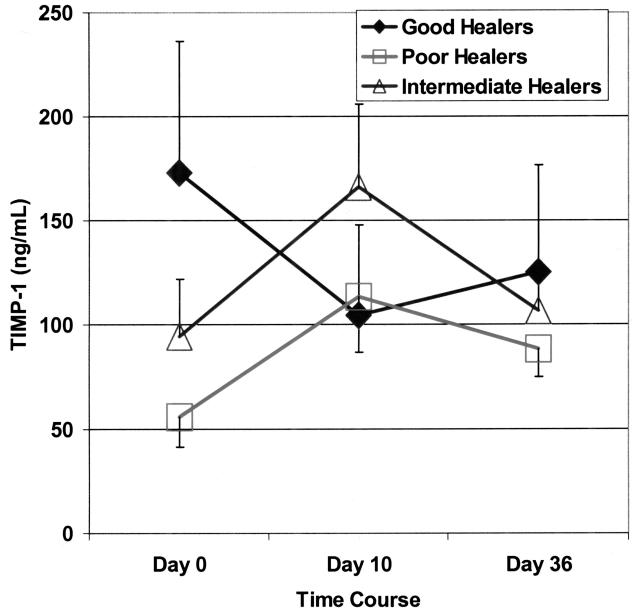


FIGURE 10. Levels of TIMP-1 in wound fluids categorized by clinical response group. Levels of TIMP-1 were significantly higher in good healers at day 0 (pretreatment) than in poor healers ($p = 0.043$). (Good healers: $n = 12$; Poor healers: $n = 8$; Intermediate healers: $n = 36$).

TIMP-1 increased in fluids on day 10 from patients who were poor or intermediate healers, and on day 36 were similar to the levels of TIMP-1 in patients who healed well.

Ratio of MMP-9/TIMP-1 as healing progressed

The ratio of the levels of MMPs and TIMPs in wound fluid samples likely reflect the actual proteolytic environment for each pressure ulcer more than levels of MMPs or TIMPs alone. Therefore, the ratio of total MMP-9 (pro + activated)/TIMP-1 was calculated for each wound fluid sample. As shown in Figure 11, the ratio of total MMP-9/TIMP-1 was significantly lower in patients whose pressure ulcers healed well at all three time points (days 0, 10, and 36) compared with patients whose ulcers healed poorly or healed intermediately. Furthermore, the ratio of MMP-9/TIMP-1 decreased significantly ($p \leq 0.05$) on days 10 and 36 compared with initial (day 0) values in the fluid samples from patients whose ulcers healed poorly. There was a similar trend in patients whose ulcers healed intermediately. The number of biopsies was too small (14 patients) for correlative analysis for the levels of MMPs and TIMPs in biopsies versus healing status.

Ratio of MMP-9/TIMP-1 on day 0 predicted extent of healing on day 36

The largest difference in ratios of MMP-9/TIMP-1 between the groups of patients occurred on day 0, where the ratio

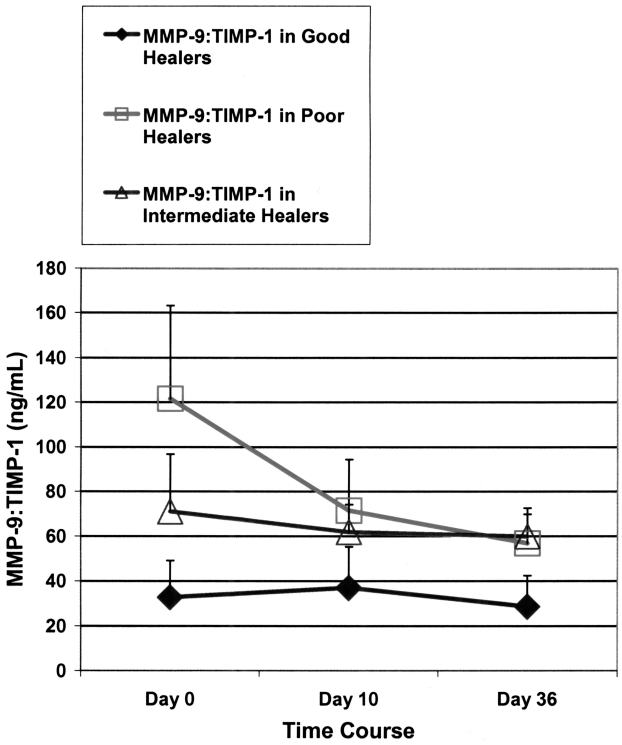


FIGURE 11. Ratio of total MMP-9/TIMP-1 in wound fluids categorized by clinical response group. Significant differences were found at day 0 between good healers and poor healers ($p = 0.047$), between good healers and intermediate healers ($p = 0.034$), and between poor healers and intermediate healers ($p = 0.029$). (Good healers: $n = 12$; Poor healers: $n = 8$; Intermediate healers: $n = 36$).

was approximately four-fold higher in patients whose pressure ulcers healed poorly at the end of the 36 week study, and approximately twofold higher in patients whose pressure ulcers healed intermediately, compared to patients who healed well ($p < 0.05$). Because the ratio of MMP-9/TIMP-1 in wound fluid was significantly correlated with the eventual outcome of healing, the initial MMP-9/TIMP-1 ratio on day 0 was a prognostic indicator of eventual clinical healing at 36 weeks.

DISCUSSION

It is important to recognize that normal wound healing requires that specific proteinases are produced at distinct locations for precise periods of time and at controlled levels.[32,33] For example, epidermal regeneration requires the production of MMP-1 by epidermal cells located at the leading edge of the migrating sheet of epidermal cells.[34] Proteinases are also necessary for removal of matrix proteins that become denatured during tissue injury, and in later stages of healing MMPs remodel the initial scar matrix, resulting in a more functional mature scar. MMPs also are required for angiogenesis. Vascular endothelial

cells secrete MMPs that degrade the basement membrane and enable new capillary loops to emerge.[35] Other proteinases are required for activation of latent growth factors and for conversion of procollagen molecules to tropocollagen molecules that can associate into collagen fibers. However, prolonged, elevated levels of proteinases appear to have detrimental effects on wound healing. Furthermore, a reduction in MMP activities in fluids collected from chronic venous ulcers was associated with the initiation of healing in the ulcers,[4] and a reduction in plasminogen activator was associated with initiation of healing in a pressure ulcer.[20] At present there are no reports correlating the rates of healing of chronic pressure ulcers with the levels of MMPs and TIMPs in fluids. This study measured levels of MMPs and TIMPs in sequential fluids and biopsies from a series of 56 patients with chronic pressure ulcers that were treated with conventional therapy or with exogenous cytokine therapies and found a strong correlation between a low ratio of MMP-9/TIMP-1 and healing of the ulcers.

Previous reports provided data supporting the hypothesis that elevated levels of cytokines and proteinases, especially MMPs, contribute to the failure of wounds to heal. For example, Tarnuzzer and Schultz[36] reported that proteinase levels (measured using Azocoll assay) in fluids from chronic wounds were significantly elevated relative to acute wound fluids. Similar results were reported by Wysocki and colleagures[6] and by Yager and colleagues[10] by gelatin zymography. Trengove et al.[4] extended these observations by showing that proteinase levels (measured by Azocoll assay) in wound fluids from 15 chronic venous ulcers significantly decreased 2 weeks after the ulcers began to heal. The present study provides the strongest data yet supporting this hypothesis by showing a strong correlation between good healing and low ratios of MMP-9/TIMP-1 in a large series of patients with chronic pressure ulcers treated in a prospective, randomized, masked study.[23] However, this study, as well as the previous studies, provides correlative data and does not directly test the hypothesis that elevated proteinases actively cause wounds to fail to heal. To answer that question, future clinical studies need to be performed to treat chronic wounds with selective proteinase inhibitors and assess correlations in healing.

Previous studies provided important information regarding proteinases and chronic wounds, yet several important questions remain. For example, it is not known if fluid samples and biopsies from chronic wounds contain similar profiles and relative levels of MMPs and TIMPs, or if data generated from analysis of wound fluids or biopsies allow a better assessment of the molecular environment of chronic wounds. Specifically, no direct comparisons of the levels of MMPs and TIMPs in fluids and biopsies collected from the same wounds have been reported. This study is the first to compare gelatin zymogram profiles and proteinase levels in both chronic wound fluids and biopsies from pressure ulcer patients. As indicated by the gelatin zymograms in Figure 3, both biopsies (panel A) and wound fluids (panel B) typically contain intense gelatinolytic bands corresponding to pro MMP-9 and activated MMP-9. In contrast, wound fluids (panel B) typically have very faint lysis bands corresponding to pro MMP-2 and activated MMP-2 compared with the intensities of pro MMP-2 and activated MMP-2 bands in biopsies (panel A). This band pattern is reinforced by the quantitative data presented in Figure 4, which show that the average levels of pro and activated MMP-2 are significantly higher than the average levels of pro and activated MMP-9 in the biopsies, while the opposite relationship is found in wound fluids. Thus, samples of wound fluids and biopsies do not have the same profiles of MMP-2 and MMP-9.

Another important distinction between MMP profiles of biopsies and wound fluid samples involves the activation of the latent forms of MMP-2 and MMP-9. Specifically, as shown in Figures 3 and 4, biopsy samples contained extremely low levels of activated MMP-9 in marked contrast to wound fluid samples, in which more than 50% of the total MMP-9 was activated. The opposite trend was observed for latent and activated MMP-2, with substantial amounts of activated MMP-2 present in biopsy samples (about 25% of total MMP-2 was activated MMP-2) and very low levels of activated MMP-2 in wound fluid samples. These data indicate that there are major differences between the MMP-2 and MMP-9 compositions of wound fluids and biopsies. The explanations for these differences are not known. However, if pro and activated MMP-2 bind more tightly to ECM components than pro and activated MMP-9, then their relative levels in biopsies should be higher than in wound fluids. Extraction of the biopsies with nonionic detergent may dissociate the pro and activated MMP-2 from the matrix proteins, which would increase their concentration in the biopsy extracts. Interestingly, immunohistochemical staining for MMP-2 tends to be substantially more intense than for MMP-9 in the ECM and fibroblasts for most biopsies (unpublished observations).

Previous studies have used different assays to measure proteinase activities in wound fluid samples. We used the Azocoll assay to measure proteinase activity in 40 wound fluid samples collected from various types of chronic wounds and in 22 samples of acute wound fluids collected from mastectomy drains.[4] The average level of proteinase activity in the chronic wound fluids was 60–71 μg/ml (median, 23 μg/ml), which was 30-fold higher

than the average level of 0.8–0.3 μg/ml in the 22 acute wound fluid samples (median, 0.8 μg/ml). The Azocoll proteinase assay measures the combined activity of all proteinases that have the ability to degrade ECM, using bovine hide as the substrate. Thus, the Azocoll assay preferentially measures gelatinase activity, but other proteinases, such as neutrophil elastase, are able to degrade the substrate. In addition, wound fluids were collected in our previous study by covering the ulcers with an occlusive dressing and allowing fluids to spontaneously accumulate for 1 hour. In this study, porous, inert, hydrophilic dextranomer beads were used to adsorb the wound fluids over a 24-hour period. Nevertheless, the combined levels of activated MMP-2 and activated MMP-9 activities reported in this current study using gelatin zymography (approximately 5 μg/ml) are relatively similar to the levels of proteolytic activity measured by Azocoll assay in our previous study (60 μg/ml).

Yager and colleagues[10] also used quantitative gelatin zymography to analyze five samples of wound fluid obtained from patients with pressure ulcers and five samples of fluids collected from patients with acute mastectomy wounds. The chronic wound fluids were collected under an occlusive dressing over a 4–6-hour period, and the average level of pro MMP-9 was 186 ± 59 ng/100 μg protein. Assuming the average protein concentration of wound fluids collected under occlusive dressings is approximately 40 mg/ml,[37] the pro MMP-9 activities would equate to 74 ± 24 μg/100 μg protein, which is similar to the level of 10 μg pro MMP-9/ml detected in our current experiments. In addition, Yager and colleagues[10] reported the average level of pro MMP-9 activity was approximately 50-fold higher than the average level of pro MMP-2 activity in the chronic wound fluids, which agrees with our finding of about eightfold higher MMP-9 activity than MMP-2 activity in the pressure ulcer wound fluids.

We previously proposed the hypothesis that the presence of elevated levels of MMPs in chronic wounds contributed to their failure to heal.[21] Analysis of the data in this study showed a statistically significant correlation between elevated levels of activated MMP-9 and poor healing, regardless of the treatment regimen that patients received. Furthermore, combining information on the levels of TIMP-1 and MMP-9 on wound fluids revealed that the ratio of the TIMP and MMP-9 also showed a statistically significant correlation between poor healing and an elevated ratio of MMP-9/TIMP-1. Thus, the data generated in this study provide strong additional support for the concept that high levels of proteolytic activity and low levels of MMP inhibitors are detrimental to healing. However, these data are merely correlative and did not provide direct evidence for a cause-and-effect relationship between high levels of proteinases and poor wound healing. Thus, there is a need to conduct the next stage of investigation of this hypothesis. This could involve a clinical study evaluating the effect of topical application of an MMP inhibitor on healing of chronic wounds. Several candidate inhibitors can be suggested, including the very potent synthetic dipeptide MMP inhibitor, Ilomostat, or the less potent but readily available antibiotic, doxycycline. In addition to its antibiotic activity, doxycycline is a competitive inhibitor of most MMPs.[38] Ilomostat and doxycycline (unpublished data) also inhibit the tumor necrosis factor-α (TNF-α)-converting enzyme, which is a metalloproteinase, and prevent TNF-α release from cultures of inflammatory cells that are stimulated by endotoxin.[39] Thus, Ilomostat or doxycycline could potentially act to reduce the levels of both the pro-inflammatory cytokine, TNF-α and the activity of MMPs if applied topically to chronic wounds.[14] The data generated in these series of experiments justify further evaluation of these new strategies and could eventually result in an extensive adjuvant therapy that could promote healing of chronic wounds.

References

1. Bennett NT, Schultz GS. Growth factors and wound healing: biochemical properties of growth factors and their receptors. Am J Surg 1993;165:728–37.
2. Bennett NT, Schultz GS. Growth factors and wound healing. Part II. Role in normal and chronic wound healing. Am J Surg 1993;166:74–81.
3. Nwomeh BC, Yager DR, Cohen IK. Physiology of the chronic wound. Clin Plast Surg 1998;25:341–56.
4. Trengove NJ, Stacey MC, Macauley S, Bennett N, Gibson J, Burslem F, Murphy G, Schultz G. Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. Wound Rep Reg 1999;7:442–52.
5. Wysocki AB, Grinnell F. Fibronectin profiles in normal and chronic wound fluid. Lab Invest 1990;63:825–31.
6. Wysocki AB, Staiano-Coico L, Grinnell F. Wound fluid from chronic leg ulcers contains elevated levels of metalloproteinases MMP-2 and MMP-9. J Invest Dermatol 1993;101:64–8.
7. Yager DR, Chen SM, Ward SI, Olutoye OO, Diegelmann RF, Cohen IK. Ability of chronic wound fluids to degrade peptide growth factors is associated with increased levels of elastase activity and diminished levels of proteinase inhibitors. Wound Rep Reg 1997;5:23–32.
8. Harris IR, Yee KC, Walters CE, Cunliffe WJ, Kearney JN, Wood EJ, Ingham E. Cytokine and protease levels in healing and non-healing chronic venous leg ulcers. Exp Dermatol 1995;4: 342–9.
9. Nwomeh BC, Liang HX, Cohen IK, Yager DR. MMP-8 is the predominant collagenase in healing wounds and nonhealing ulcers. J Surg Res 1999;81:189–95.
10. Yager DR, Zhang LY, Liang HX, Diegelmann RF, Cohen IK. Wound fluids from human pressure ulcers contain elevated matrix metalloproteinase levels and activity compared to surgical wound fluids. J Invest Dermatol 1996;107:743–8.
11. Trengove NJ, Bielefeldt-Ohmann H, Stacey MC. Mitogenic activity and cytokine levels in non-healing and healing chronic leg ulcers. Wound Rep Reg 2000;8:13–25.
12. Bucalo B, Eaglstein WH, Falanga V. Inhibition of cell proliferation by chronic wound fluid. Wound Rep Reg 1993;1:181–6.

13. Wlaschek M, Pes D, Achterberg V, Meyer-Ingold W, Scharfetter-Kochanek K. Protease inhibitors protect growth factor activity in chronic wounds. Br J Dermatol 1997;137:646–7.
14. Weckroth M, Vaheri A, Lauharanta J, Sorsa T, Konttinen YT. Matrix metalloproteinases, gelatinase, and collagenase in chronic leg ulcers. J Invest Dermatol 1996;106:1119–24.
15. Rao CN, Ladin DA, Liu YY, Chilukuri K, Hou ZZ, Woodley DT. a1-Antitrypsin is degraded and non-functional in chronic wounds but intact and functional in acute wounds: the inhibitor protects fibronectin from degradation by chronic wound fluid enzymes. J Invest Dermatol 1995;105:572–8.
16. Bullen EC, Longaker MT, Updike DL, Benton R, Ladin D, Hou Z. Tissue inhibitor of metalloproteinases-1 is decreased and activated gelatinases are increased in chronic wounds. J Invest Dermatol 1995;104:236–40.
17. Rogers AA, Burnett S, Moore JC, Shakespeare PG, Chen WYJ. Involvement of proeolytic enzymes, plasminogen activators and matrix metalloproteinases in the pathophysiology of pressure ulcers. Wound Rep Reg 1995;3:273–83.
18. Barone EJ, Yager DR, Pozez AL, Olutoye OO, Crossland MC, Diegelmann RF, Cohen IK. Interleukin-1alpha and collagenase activity are elevated in chronic wounds. Plastic Reconstr Surg 1998;102:1023–7.
19. Tarlton JF, Vickery CJ, Leaper DJ, Bailey AJ. Postsurgical wound progression monitored by temporal changes in the expession of matrix metalloproteinase-9. Br J Dermatol 1997;137:506–16.
20. Wysocki AB, Kusakabe AO, Chang S, Tuan TL. Temporal expression of urokinase plasminogen activator, plasminogen activator inhibitor and gelatinase-B in chronic wound fluid switches from a chronic to acute wound profile with progression to healing. Wound Rep Reg 1999;7:154–65.
21. Mast BA, Schultz GS. Interactions of cytokines, growth factors, and proteases in acute and chronic wounds. Wound Rep Reg 1996;4:411–20.
22. Yager DR, Nwomeh BC. The proteolytic environment of chronic wounds. Wound Rep Reg 1999;7:433–41.
23. Robson MC, Hill DP, Smith PD, Wang X, Meyer-Siegler K, Ko F, VandeBerg JS, Payne WG, Ochs D, Robson LE. Sequential cytokine therapy for pressure ulcers: clinical and mechanistic response. Ann Surg 2000;231:600–11.
24. Resch CS, Kerner E, Robson MC, Heggers JP, Scherer M, Boertman JA, Schileru R. Pressure sore volume measurement. A technique to document and record wound healing. J Am Geriatr Soc 1988;36: 444–6.
25. Cooper DM, Robson MC. In the evaluation of potential wound healing agents a placebo arm is not nontreatment. Wound Rep Reg 1996;4:A128. [Abstract].
26. Matsuoka J, Grotendorst GR. Two peptides related to platelet-derived growth factor are present in human wound fluid. Proc Natl Acad Sci USA 1989;86:4416–20.
27. Fini ME, Girard MT. The pattern of metalloproteinase expression by corneal fibroblasts is altered by passage in cell culture. J Cell Sci 1990;97(Pt. 2):373–83.
28. Hibbs MS, Hasty KA, Seyer JM, Kang AH, Mainardi CL. Biochemical and immunological characterization of the secreted forms of human neutrophil gelatinase. J Biol Chem 1985;260:2493–500.
29. Kleiner DE, Stetler-Stevenson WG. Quantitative zymography: detection of picogram quantities of gelatinases. Anal Biochem 1994;218:325–9.
30. Todor DR, Lewis I, Bruno G, Chyatte D. Identification of a serum gelatinase associated with the occurrence of cerebral aneurysms as pro-matrix metalloproteinase-2. Stroke 1998;29:1580–3.
31. Mauviel A. Cytokine regulation of metalloproteinase gene expression. J Cell Biochem 1993;53:288–95.
32. Parks WC. Matrix metalloproteinases in repair. Wound Rep Reg 1999;7:423–32.
33. Barrick B, Campbell EJ, Owen CA. Leukocyte proteinases in wound healing: roles in physiologic and pathologic processes. Wound Rep Reg 1999;7:410–22.
34. Parks WC. The production, role, and regulation of matrix metalloproteinsases in the healing epidermis. Wounds 1995;7:23A–37A.
35. Fisher C, Gilbertson-Beadling S, Powers EA, Petzold G, Poorman R, Mitchell MA. Interstitial collagenase is required for angiogenesis in vitro. Dev Biol 1994;162:499–510.
36. Tarnuzzer RW, Schultz GS. Biochemical analysis of acute and chronic wound environments. Wound Rep Reg 1996;4:321–5.
37. Trengove NJ, Langton SR, Stacey MC. Biochemical analysis of wound fluid from nonhealing and healing chronic leg ulcers. Wound Rep Reg 1996;4:234–9.
38. Smith GN Jr, Mickler EA, Hasty KA, Brandt KD. Specificity of inhibition of matrix metalloproteinase activity by doxycycline: relationship to structure of the enzyme. Arthritis Rheum 1999;42:1140–6.
39. Solorzano CC, Ksontini R, Pruitt JH, Auffenberg T, Tannahill C, Galardy RE, Schultz GP, MacKay SL, Copeland EM III, Moldawer LL. A matrix metalloproteinase inhibitor prevents processing of tumor necrosis factor alpha (TNF alpha) and abrogates endotoxin-induced lethality. Shock 1997;7:427–31.